AI Image Classifier using CNN

Professional Project Documentation (GitHub Ready)

1. Project Overview

This project is a full-stack AI Image Classifier based on Convolutional Neural Networks (CNN). The application allows users to upload images and classify them into predefined categories such as Dog and Cat. The goal of this project is to help learners understand how deep learning models are trained, deployed, and integrated into a real-world web application.

2. Problem Statement

Traditional image classification systems require manual feature extraction and are not scalable. This project solves the problem by using CNN-based deep learning models that automatically learn features from images and provide accurate predictions.

3. Objectives

- Build a CNN model for image classification
- Create a user-friendly frontend for image upload
- Integrate AI model with a backend API
- Provide an end-to-end deployable solution

4. Technologies & Tools Used

Frontend: React.js, HTML, CSS, JavaScript
Backend: Python, Flask / FastAPI
AI / ML: TensorFlow, Keras, CNN
Others: Git, GitHub, VS Code

5. System Architecture

1. User uploads an image from the frontend
2. Image is sent to backend API
3. Backend preprocesses the image
4. CNN model predicts the class
5. Result is sent back to frontend

6. CNN Working Model (Step-by-Step)

Step 1: Image Input
User uploads an image which is resized and normalized.

Step 2: Convolution Layer
Extracts important features like edges and shapes.

Step 3: Pooling Layer
Reduces image size while keeping important features.

Step 4: Fully Connected Layer
Learns high-level patterns and relationships.

Step 5: Output Layer
Uses Softmax/Sigmoid to give final prediction.

7. Model Training Process

The dataset is divided into training and validation sets. Images are augmented to improve generalization. The model is trained using backpropagation and optimized using Adam optimizer.

8. Folder Structure

/frontend
/backend
/model
/dataset
README.md

9. How to Run the Project

1. Clone the GitHub repository
2. Install frontend dependencies
3. Install backend requirements
4. Run backend server
5. Start frontend application

10. Use Cases

- Learning CNN fundamentals
- AI/ML portfolio project
- Academic demonstrations
- Beginner-friendly practice project

11. Future Enhancements

- Add more image categories
- Improve model accuracy
- Deploy on cloud platform
- Add user authentication

12. Conclusion

This AI Image Classifier CNN project demonstrates a complete machine learning lifecycle from data preparation to deployment. It is designed for learners who want hands-on experience with real-world AI systems.